

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 The phenotype of an organism is affected by a number of factors.

(a) State what is meant by the term **phenotype**.

(1)

.....

.....

.....

(b) Coat colour in rabbits is determined by multiple alleles.

The table gives some information about coat colour in rabbits.

Type of rabbit	Coat colour of rabbit	
Black 	black all over	CC
Chinchilla 	grey all over	$c^{ch}c^{ch}$
Himalayan 	white body black ears, face, feet and tail	c^hc^h
Albino 	white all over	cc

(i) Complete this table by writing a suitable heading for the right-hand column.

(1)

- (ii) Which row of the table gives the correct number of genes and alleles for coat colour in these rabbits?

(1)

	Number of genes for coat colour	Number of alleles for coat colour
☐ A	1	1
☐ B	1	4
☐ C	4	1
☐ D	4	4

- (c) Height is one phenotype of an elephant.

The photograph shows an African elephant.



Source: Caroline Wilcox

Male African elephants range in height from 3.2 m to 4.0 m.

Female African elephants range in height from 2.2 m to 2.6 m.

- (i) Which row of the table names the types of graph that should be drawn to show sex and height variation in a population of African elephants?

(1)

	Sex	Height
☐ A	bar chart	bar chart
☐ B	bar chart	histogram
☐ C	histogram	bar chart
☐ D	histogram	histogram

- (ii) Calculate how many times bigger the male African elephant is than the female African elephant.

(2)

Answer

(Total for Question 1 = 6 marks)

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